

IMPERIAL

Atoms and Molecules in Tweezers

State of the Experiment

Thomas Walker
August 5, 2026

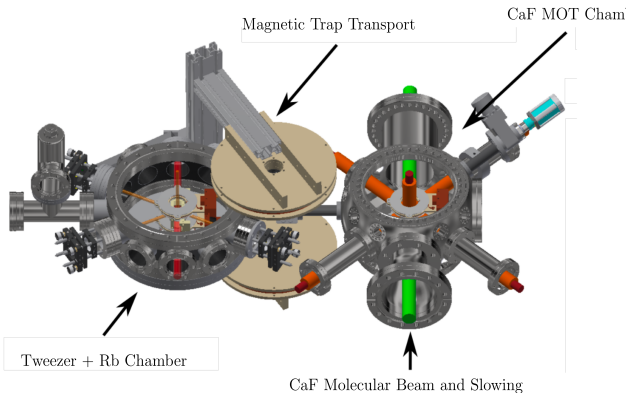
The Chamber(s)

Trapping CaF Molecules

- CaF blue MOT $\sim 2.5 - 3.0 \times 10^4$ molecules at ~ 60 μ K.
- Optical pumping, magnetic trapping of CaF and transport to the tweezer chamber.
- Transfer into a blue MOT in the tweezer chamber.
- Load CaF into 852 nm tweezer array from blue MOT.

Rydberg Excitation

- **780 nm + 480 nm** - two-photon Rydberg excitation to 60s.
- **297 nm** - direct single-photon Rydberg excitation to 59p.

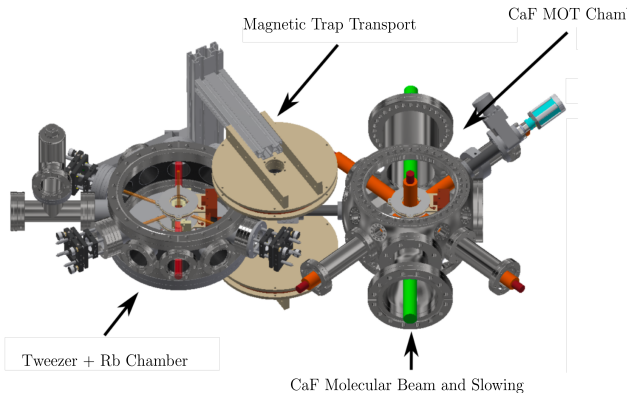


Second Wavelength Tweezer Array

- Set up a second high-speed SLM for generating an array which is selectively attractive to CaF.
- Small site spacing of $\sim 1.0 \mu\text{m}$ between CaF and Rb without interference.
- Species selective trapping.

Electric Field Control

- Replace internal mount - install Faraday shielding and electrodes.
- Tuning the Rb Rydberg energy separation to match the CaF rotational energy separation for a resonant dipole interaction.



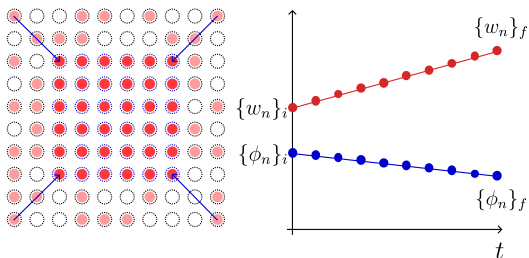
High-Speed Parallel Rearrangement

Tweezer Array Rearrangement

- Stochastic loading of Rb (~55%) and CaF (~30%).
- High-speed parallel rearrangement for **near-deterministic defect-free arrays** in arbitrary geometries.
- Pre-calculate initial and final array phase masks using full phase-retrieval algorithm (5 sec).
- Intermediate phase masks calculated by **linearly interpolating** trap weights, phases, positions.
- Map occupied trap sites to final trap sites using an **efficient sorting algorithm** (e.g. Hungarian, Jonker-Volgenant algorithms).

$$\Phi(u, v) = \arg \left(\sum_n^N w_n e^{i(kx_n u + ky_n v + \phi_n)} \right)$$

$$\begin{aligned} \{x_n\}_i &\rightarrow \{x_n\}_f \\ \{y_n\}_i &\rightarrow \{y_n\}_f \end{aligned}$$

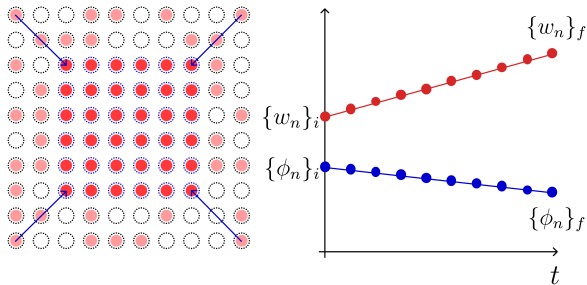


Tweezer Array Rearrangement

1. Load and image tweezer array (20 ms exposure).
2. Process image and extract occupancy mask of array (2 ms).
3. Map occupied trap sites to final trap sites using JV sorting algorithm (<1ms).
4. Switch off unoccupied and excess trap sites.
5. Interpolate, calculate next phase mask + upload phase mask to SLM (in parallel, ~1.8 ms per frame, typ. 20 ms total rearrangement time).

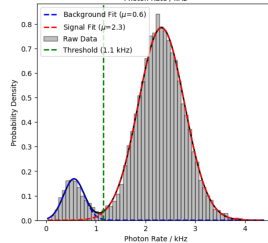
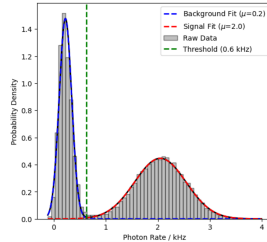
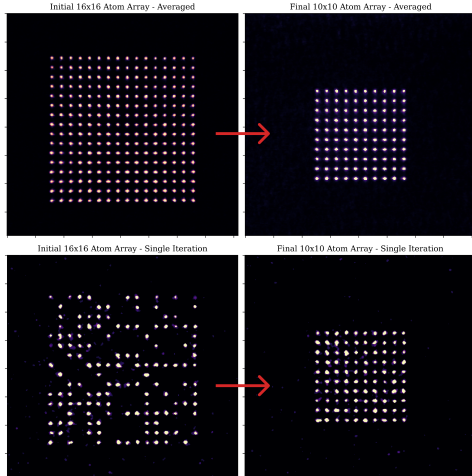
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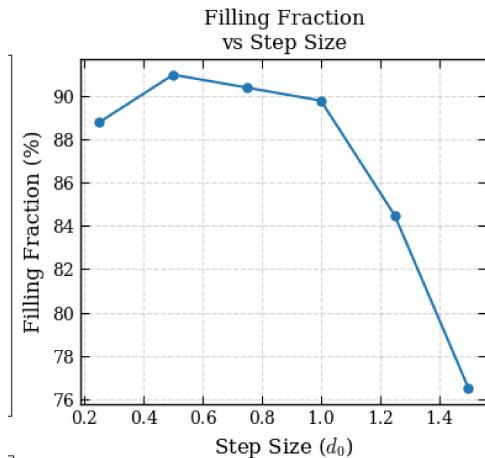
Tweezer Array Rearrangement

Rearrangement Performance



Tweezer Array Rearrangement

Filling Fraction vs Step Size



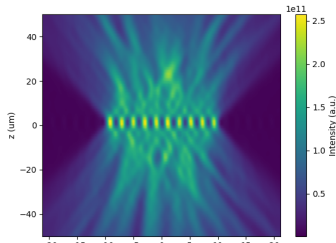
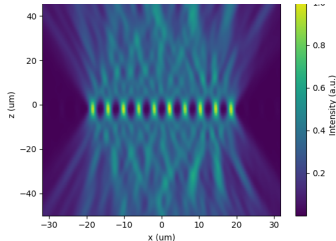
- Max. step size $\sim 1.0 \mu\text{m}$
- Rearrangement 'speed' = 1.88 ms per frame
- 12.2 frames per sequence, 22.9 ms total rearrangement time
- Loss dominated by lifetime + imaging loss.
- Rearrangement 'efficiency' $\sim 98\%$.

Dense Arrays

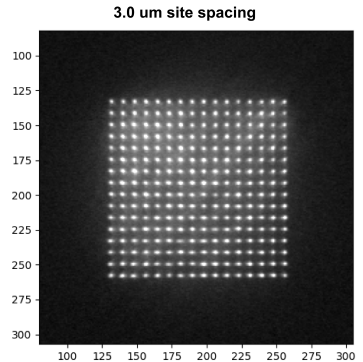
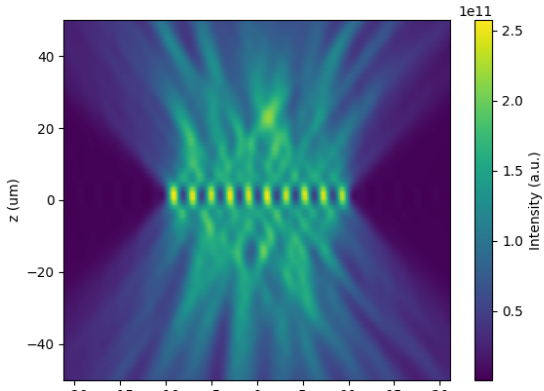
For two-qubit hybrid gates need small trap site separation $\sim 2.0 \mu\text{m}$ for high-fidelity operations.

Challenges:

- Interference of tweezer light, deep off-plane traps, reduces detection fidelity.
- Interference between neighbouring trap sites, distorts trap potential.
- Rearrangement linear interpolation method breaks down for small site spacing.



Dense Arrays



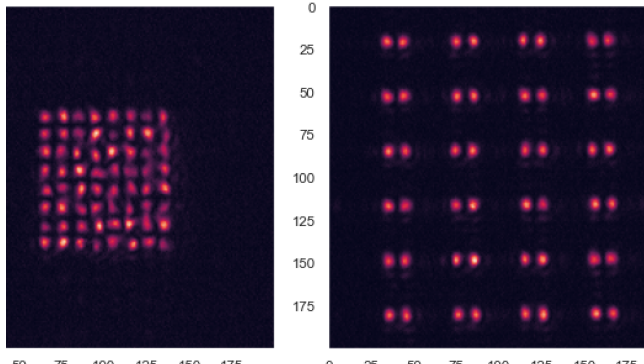
Dense Arrays

For two-qubit hybrid gates need small trap site separation $\sim 2.0 \mu\text{m}$ for high-fidelity operations.

Challenges:

- Interference of tweezer light, deep off-plane traps which can efficiently trap, reduces detection fidelity.
- Interference between neighbouring trap sites, distorts trap potential.
- Rearrangement linear interpolation method breaks down for small site spacing.

2.0 μm site spacing - CCD image of array



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Thank you

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